

Business Problem

"Students are working hard — but in the wrong direction."

- High learning effort vs low job acquisition
- Focus on trendy skills (ML) vs foundational (SQL)
- Massive drop-off (55%) post-application

Proposed Interventions

- ✓

Intensive SQL Bootcamp:

Address 28% critical market shortage
- ✓

Guided Learning Paths:

Stop undirected study causing reverse causality
- ✓

Structured Mentorship:

Primary differentiator for successful candidates
- ✓

Deprioritize ML:

Currently oversupplied by 22% for entry-level

Analysis Methodology & Key Findings



Technology Stack

- Python (Pandas/NumPy)

n8n Automation
- Claude AI API

Scikit-learn

Playwright
- Power BI / Excel

Development Phases

- Phase 0 & 1

Data Prep & Descriptive Analysis
- Phase 2

Core Gap Analysis (Demand vs Supply)
- Phase 3 & 4

Effort/Results Modeling & Stat Testing
- Phase 5

KMeans Segmentation (k=4)

Project Deliverables

- ✓

Cleaned Datasets
- ✓

Gap Analysis Results
- ✓

Stat Test & ML Profiles
- ✓

Interactive Dashboard
- ✓

Strategy Roadmap